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Subject: Programming with Java	
ASSIGNMENT No : 3	
TITLE : Method Overloading and Static Members	

Problem Statement

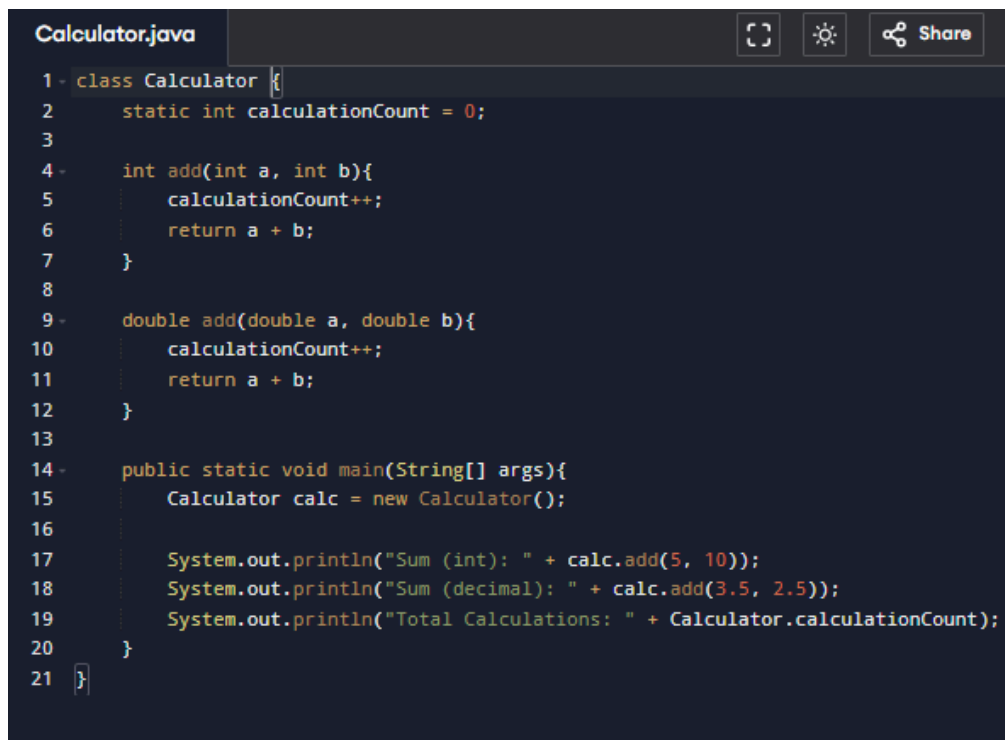
Develop a Java application demonstrating method overloading and the use of static variables, methods in class design.

Learning Objectives

To implement method overloading and use static variables and methods in Java programs.

Programme

1. Develop a Calculator program using overloaded methods for addition of integers and decimals. Use a static variable to count calculations.



```
Calculator.java
1 class Calculator {
2     static int calculationCount = 0;
3
4     int add(int a, int b){
5         calculationCount++;
6         return a + b;
7     }
8
9     double add(double a, double b){
10        calculationCount++;
11        return a + b;
12    }
13
14    public static void main(String[] args){
15        Calculator calc = new Calculator();
16
17        System.out.println("Sum (int): " + calc.add(5, 10));
18        System.out.println("Sum (decimal): " + calc.add(3.5, 2.5));
19        System.out.println("Total Calculations: " + Calculator.calculationCount);
20    }
21 }
```

2. Develop a Restaurant Billing Application where overloaded methods calculate bills for dine-in, takeaway, and delivery orders, while static variables track total orders.

```
RestaurantBilling.java    Share  Run

1 class RestaurantBilling {
2     static int totalOrders = 0;
3
4     double calculateBill(double foodAmount, double taxPercentage) {
5         totalOrders++;
6         return foodAmount + (foodAmount * taxPercentage / 100);
7     }
8
9     double calculateBill(double foodAmount, double taxPercentage, double packagingCharge) {
10        totalOrders++;
11        return foodAmount + (foodAmount * taxPercentage / 100) + packagingCharge;
12    }
13
14    double calculateBill(double foodAmount, double taxPercentage, double packagingCharge,
15        double deliveryFee) {
16        totalOrders++;
17        return foodAmount + (foodAmount * taxPercentage / 100) + packagingCharge + deliveryFee;
18    }
19
20    public static void main(String[] args) {
21        RestaurantBilling bill = new RestaurantBilling();
22
23        System.out.println("Dine-in Bill: Rs" + bill.calculateBill(500.0, 50.0));
24        System.out.println("Takeaway Bill: Rs" + bill.calculateBill(500.0, 50.0, 20.0));
25        System.out.println("Delivery Bill: Rs" + bill.calculateBill(500.0, 50.0, 20.0, 50.0));
26        System.out.println("Total Orders Placed: " + RestaurantBilling.totalOrders);
27    }
```

Output

Program 1 output:

```
Output

Sum (int): 15
Sum (decimal): 6.0
Total Calculations: 2

=== Code Execution Successful ===
```

Program 2 output:

```
Output
Dine-in Bill: Rs750.0
Takeaway Bill: Rs770.0
Delivery Bill: Rs820.0
Total Orders Placed: 3

=== Code Execution Successful ===
```

Conclusion

These programs demonstrate method overloading and static variable usage in Java. The Calculator program shows how method overloading allows the same method name to handle different data types (integers and decimals), while the Restaurant Billing application uses overloading to process different order types with varying fee parameters. In both applications, static variables serve as global counters to track overall usage (total calculations and total orders) across instances.